

Data Scientist - Case Study

Please read the following instructions carefully for a case study that each candidate will need to prepare for the final interview.

Instructions:

- (1) You will have 30 minutes to present your findings on some questions about a dataset regarding red wine quality (dataset available online in kaggle).
- (2) Download the .CSV file on <https://www.kaggle.com/datasets> on “Red Wine Dataset” as per snapshot below:



 Dataset

Red Wine Dataset

by piyushgoyal443

5 years ago • 30 kB •  75

[Red Wine Dataset](#)

(Note: You will need to have some Google account to be able to download files from kaggle)

- (3) A full explanation of the data and all elements presented in the data is given on the website, so please read carefully for you to understand it. For further reference, the citation underlying this dataset has is available on kaggle. But only necessary to take note of the abstract [*Cortez et al., 2009*]
- (4) Please prepare your presentation and send it off to mike@unifi.credit by EOBD on the day before your interview. If your file(s) are ≥ 20 MB, please place in a google drive or similar and send link. These files will be ready when you pitch up for your interview. (Where applicable, also include all working files used as part of your final analysis / presentation).
- (5) The presentation should contain all answers to questions provided below and where applicable should give the most applicable output explaining your answers.
- (6) You are required to use only Microsoft products to present your findings (Excel, Word, Powerpoint).
- (7) You may use any additional analytical tool (Excel, Python, SAS, R etc.) to do your analysis

Questions that will need to be answered:

- (1) Using a cross-correlation matrix, please explain if there are intercorrelated variables in your dataset.
- (2) Generate appropriate output explaining the distributions of the underlying variables used in the dataset. Also explain various properties of your data elements.
- (3) Considering the problem presented by the author, define your model's explanatory and response variables, and give your findings of what potential type of model fits you've considered.
- (4) Do an appropriate test to see which data elements are normally distributed or not
- (5) Were any transformations / variable reduction techniques required on any of the data elements or as input to the model?
- (6) Fit an appropriate model and explain the following:
 - a. How you've approached your model and variable selection criteria
 - b. How good of a fit was the model
 - c. Your model production "formula"
- (7) In your own words, how can we use a learning algorithm to identify "good" vs. "bad" quality red wine, particularly since techniques may have changed since 2009 when this was first examined